```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

```
df=pd.read_csv('/content/HR Data (1).csv')
df
```

	Attrition	Business Travel	CF_age band	CF_attrition label	Department	Education Field	emp no	Employee Number	Gender	Job Role	...	Performance Rating	R S
0	Yes	Travel_Rarely	35 - 44	Ex-Employees	Sales	Life Sciences	STAFF-1	1	Female	Sales Executive	...	3	
1	No	Travel_Frequently	45 - 54	Current Employees	R&D	Life Sciences	STAFF-2	2	Male	Research Scientist	...	4	
2	Yes	Travel_Rarely	35 - 44	Ex-Employees	R&D	Other	STAFF-4	4	Male	Laboratory Technician	...	3	
3	No	Travel_Frequently	25 - 34	Current Employees	R&D	Life Sciences	STAFF-5	5	Female	Research Scientist	...	3	
4	No	Travel_Rarely	25 - 34	Current Employees	R&D	Medical	STAFF-7	7	Male	Laboratory Technician	...	3	
...	...	...	...	...	...	...	...	...	...	...	...	...	
1465	Yes	Non-Travel	25 - 34	Ex-Employees	R&D	Technical Degree	STAFF-1905	1905	Male	Research Scientist	...	4	
1466	Yes	Travel_Frequently	25 - 34	Ex-Employees	R&D	Life Sciences	STAFF-1868	1868	Male	Research Scientist	...	4	
1467	Yes	Travel_Frequently	35 - 44	Ex-Employees	Sales	Other	STAFF-1667	1667	Male	Sales Executive	...	4	
1468	Yes	Travel_Rarely	Under 25	Ex-Employees	R&D	Life Sciences	STAFF-1878	1878	Male	Research Scientist	...	4	
1469	Yes	Travel_Rarely	Under 25	Ex-Employees	Sales	Life Sciences	STAFF-1702	1702	Male	Sales Representative	...	4	

1470 rows × 41 columns

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1470 entries, 0 to 1469
Data columns (total 41 columns):
 #   Column                   Non-Null Count  Dtype
---  ------                   --------------  -----
 0   Attrition                1470 non-null   object
 1   Business Travel          1470 non-null   object
 2   CF_age band              1470 non-null   object
 3   CF_attrition label       1470 non-null   object
 4   Department               1470 non-null   object
 5   Education Field          1470 non-null   object
 6   emp no                   1470 non-null   object
 7   Employee Number          1470 non-null   int64
 8   Gender                   1470 non-null   object
 9   Job Role                 1470 non-null   object
 10  Marital Status           1470 non-null   object
 11  Over Time                1470 non-null   object
 12  Over18                   1470 non-null   object
 13  Training Times Last Year 1470 non-null   int64
 14  -2                       1470 non-null   int64
 15  0                        1470 non-null   int64
 16  Age                      1470 non-null   int64
 17  CF_current Employee      1470 non-null   int64
 18  Daily Rate               1470 non-null   int64
 19  Distance From Home       1470 non-null   int64
 20  Education                1470 non-null   object
 21  Employee Count           1470 non-null   int64
 22  Environment Satisfaction 1470 non-null   int64
 23  Hourly Rate              1470 non-null   int64
 24  Job Involvement          1470 non-null   int64
 25  Job Level                1470 non-null   int64
 26  Job Satisfaction         1470 non-null   int64
```

```
 27  Monthly Income             1470 non-null   int64
 28  Monthly Rate               1470 non-null   int64
 29  Num Companies Worked       1470 non-null   int64
 30  Percent Salary Hike        1470 non-null   int64
 31  Performance Rating         1470 non-null   int64
 32  Relationship Satisfaction  1470 non-null   int64
 33  Standard Hours             1470 non-null   int64
 34  Stock Option Level         1470 non-null   int64
 35  Total Working Years        1470 non-null   int64
 36  Work Life Balance          1470 non-null   int64
 37  Years At Company           1470 non-null   int64
 38  Years In Current Role      1470 non-null   int64
 39  Years Since Last Promotion 1470 non-null   int64
 40  Years With Curr Manager    1470 non-null   int64
dtypes: int64(28), object(13)
memory usage: 471.0+ KB
```

```
df.isnull().sum()
```

	0
Attrition	0
Business Travel	0
CF_age band	0
CF_attrition label	0
Department	0
Education Field	0
emp no	0
Employee Number	0
Gender	0
Job Role	0
Marital Status	0
Over Time	0
Over18	0
Training Times Last Year	0
-2	0
0	0
Age	0
CF_current Employee	0
Daily Rate	0
Distance From Home	0
Education	0
Employee Count	0
Environment Satisfaction	0
Hourly Rate	0
Job Involvement	0
Job Level	0
Job Satisfaction	0
Monthly Income	0
Monthly Rate	0
Num Companies Worked	0
Percent Salary Hike	0
Relationship Satisfaction	0
Standard Hours	0

```
df.duplicated().sum()  # To check duplicate values
```

```
np.int64(0)
```

```
df.drop('0',axis=1,inplace=True) #Droping unnecessary columns
```

```
df.drop('-2',axis=1,inplace=True)
```

Years At Company 0

Years Since Last Promotion 0

Years With Curr Manager 0

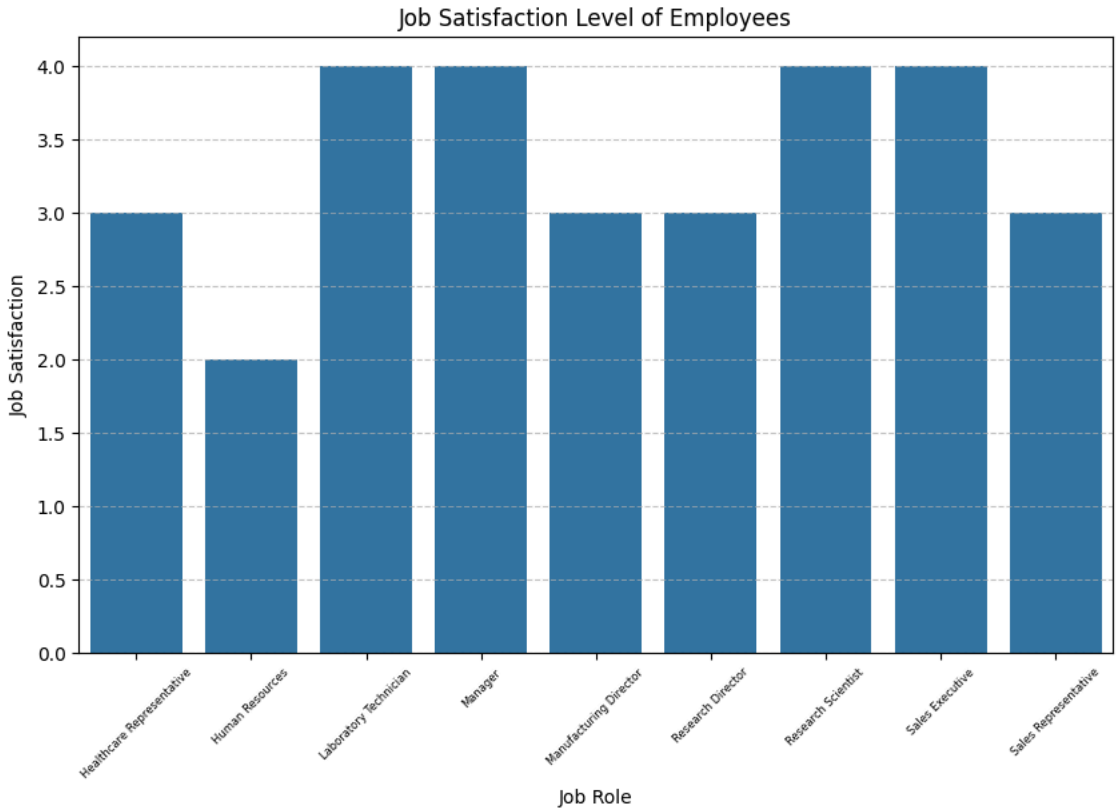
dtype: int64

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1470 entries, 0 to 1469
Data columns (total 39 columns):
 #   Column                      Non-Null Count  Dtype
---  ------                      --------------  -----
 0   Attrition                   1470 non-null   object
 1   Business Travel             1470 non-null   object
 2   CF_age band                 1470 non-null   object
 3   CF_attrition label          1470 non-null   object
 4   Department                  1470 non-null   object
 5   Education Field             1470 non-null   object
 6   emp no                      1470 non-null   object
 7   Employee Number             1470 non-null   int64
 8   Gender                      1470 non-null   object
 9   Job Role                    1470 non-null   object
 10  Marital Status              1470 non-null   object
 11  Over Time                   1470 non-null   object
 12  Over18                      1470 non-null   object
 13  Training Times Last Year    1470 non-null   int64
 14  Age                         1470 non-null   int64
 15  CF_current Employee         1470 non-null   int64
 16  Daily Rate                  1470 non-null   int64
 17  Distance From Home          1470 non-null   int64
 18  Education                   1470 non-null   object
 19  Employee Count              1470 non-null   int64
 20  Environment Satisfaction    1470 non-null   int64
 21  Hourly Rate                 1470 non-null   int64
 22  Job Involvement             1470 non-null   int64
 23  Job Level                   1470 non-null   int64
 24  Job Satisfaction            1470 non-null   int64
 25  Monthly Income              1470 non-null   int64
 26  Monthly Rate                1470 non-null   int64
 27  Num Companies Worked        1470 non-null   int64
 28  Percent Salary Hike         1470 non-null   int64
 29  Performance Rating          1470 non-null   int64
 30  Relationship Satisfaction   1470 non-null   int64
 31  Standard Hours              1470 non-null   int64
 32  Stock Option Level          1470 non-null   int64
 33  Total Working Years         1470 non-null   int64
 34  Work Life Balance           1470 non-null   int64
 35  Years At Company            1470 non-null   int64
 36  Years In Current Role       1470 non-null   int64
 37  Years Since Last Promotion  1470 non-null   int64
 38  Years With Curr Manager     1470 non-null   int64
dtypes: int64(26), object(13)
memory usage: 448.0+ KB
```

1.On a scale of 1-10, how satisfied are you with your current role and responsibilities?

```
mode_role=df['Job Satisfaction'].groupby(df['Job Role']).apply(lambda x: x.mode().iloc[0])

mode_df=mode_role.reset_index()
mode_df.columns=['Job Role','Job Satisfaction']

plt.figure(figsize=(10,6))
sns.barplot(data=mode_df,x='Job Role',y='Job Satisfaction')
plt.title("Job Satisfaction Level of Employees")
plt.xticks(rotation=45, fontsize=6)
plt.grid(axis='y', linestyle='--', alpha=0.7)
plt.show()
```



Since, majority of the employees give 4 and 3 ratings. we can conclude that the employees are satisfied with their current role and responsibilities.

2. Do you feel your work is recognized and appreciated by your manager? (Yes/No)

```
df['Relationship Satisfaction'].value_counts()
```

Relationship Satisfaction	count
3	459
4	432
2	303
1	276

dtype: int64

```
sns.countplot(x='Relationship Satisfaction', data=df)
plt.title('Employees Feeling Recognized by Manager')
plt.xlabel('Satisfaction Level')
plt.ylabel('Number of Employees')
plt.show()
```

"The analysis reveals how employees perceive recognition and appreciation from their managers, with higher satisfaction levels indicating a healthier workplace environment.

3.How would you rate the work-life balance provided by the company? (Poor – Excellent)

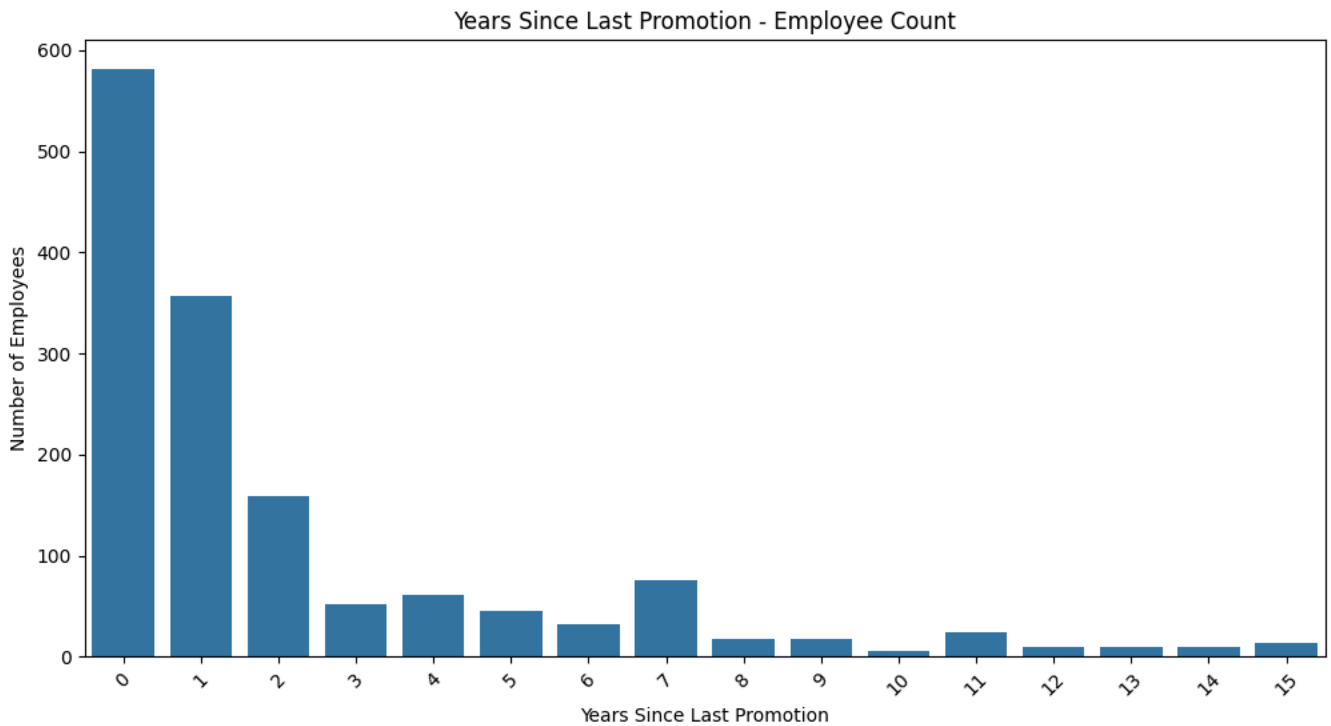
```
# Count the responses
balance_counts = df["WorkLifeBalance_Label"].value_counts().reindex(["Poor", "Average", "Good", "Excellent"])

# Plot pie chart
plt.figure(figsize=(7, 4))
plt.pie(balance_counts, labels=balance_counts.index, autopct='%1.1f%%', startangle=140, colors=sns.color_palette("bright"))
plt.title("3. How would you rate the work-life balance provided by the company?")
plt.axis('equal')  # Equal aspect ratio makes the pie chart circular
plt.show()
```

```
---------------------------------------------------------------------------
KeyError                                  Traceback (most recent call last)
/usr/local/lib/python3.11/dist-packages/pandas/core/indexes/base.py in get_loc(self, key)
   3804         try:
-> 3805             return self._engine.get_loc(casted_key)
   3806         except KeyError as err:

index.pyx in pandas._libs.index.IndexEngine.get_loc()

index.pyx in pandas._libs.index.IndexEngine.get_loc()

pandas/_libs/hashtable_class_helper.pxi in pandas._libs.hashtable.PyObjectHashTable.get_item()

pandas/_libs/hashtable_class_helper.pxi in pandas._libs.hashtable.PyObjectHashTable.get_item()

KeyError: 'WorkLifeBalance_Label'

The above exception was the direct cause of the following exception:

KeyError                                  Traceback (most recent call last)
                                 2 frames
/usr/local/lib/python3.11/dist-packages/pandas/core/indexes/base.py in get_loc(self, key)
   3810             ):
   3811                 raise InvalidIndexError(key)
-> 3812             raise KeyError(key) from err
   3813         except TypeError:
   3814             # If we have a listlike key, _check_indexing_error will raise

KeyError: 'WorkLifeBalance_Label'
```

From the above graph, we can conclude that majority of employees rated "Good" work life balance.

4.Do you see opportunities for growth and development in this organization? (Yes/No)

```
plt.figure(figsize=(12, 6))
sns.countplot(data=df, x='Years Since Last Promotion', order=sorted(df['Years Since Last Promotion'].unique()))
plt.title('Years Since Last Promotion - Employee Count')
plt.xlabel('Years Since Last Promotion')
plt.ylabel('Number of Employees')
plt.xticks(rotation=45)
plt.show()
```



"Most employees who were recently promoted (0–2 years) reported higher growth opportunities (Yes), while employees with longer gaps since their last promotion leaned more towards 'No

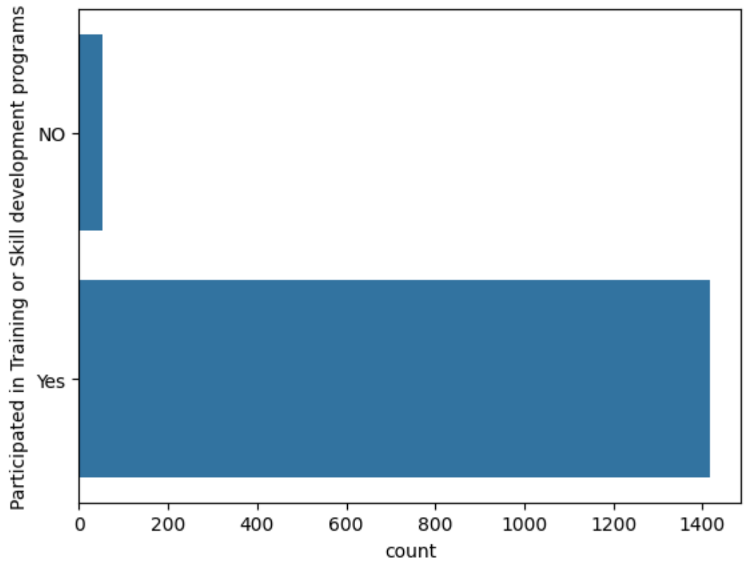
5.Have you participated in any training or skill development programs in the last 6 months? (Yes/No)

```
training_df=df['Training Times Last Year'].apply(lambda x:'Yes' if x!=0 else 'NO')
training_df.value_counts()
```

Training Times Last Year	count
Yes	1416
NO	54

dtype: int64

```
sns.countplot(data=training_df)
plt.ylabel("Participated in Training or Skill development programs")
plt.xlabel("count")
plt.show()
```



YES! Many employees are participated in any training or skill development programs

6. How does Distance From Home relate to Attrition?(Are people leaving because of long commutes?)

```
from sklearn.preprocessing import LabelEncoder
le=LabelEncoder()
df['Attrition']=le.fit_transform(df['Attrition'])
```

```
corr_matrix=df['Distance From Home'].corr(df['Attrition'])
corr_matrix
```

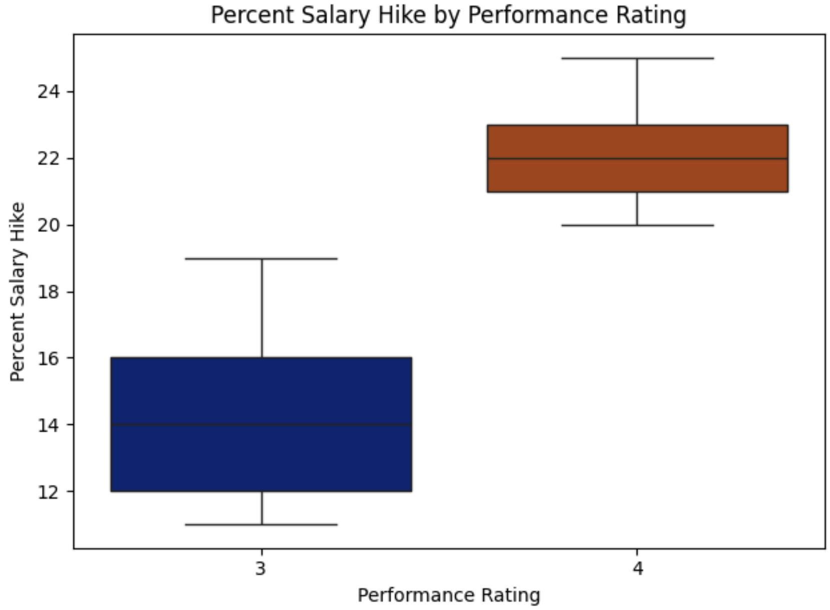
```
np.float64(0.0779235829557036)
```

We got positive correlation between Distance from home and Attrition.Therefore we can say that if distance from home is more then there is more chances for employees to leave the company.

7. Does Performance Rating align with Percent Salary Hike?(Are high performers getting paid fairly?)

```
sns.boxplot(x='Performance Rating',y='Percent Salary Hike',data=df,palette="dark")
plt.title('Percent Salary Hike by Performance Rating')
plt.xlabel('Performance Rating')
plt.ylabel('Percent Salary Hike')
plt.tight_layout()
plt.show()
```

```
<ipython-input-138-d298af271aa5>:1: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend

  sns.boxplot(x='Performance Rating',y='Percent Salary Hike',data=df,palette="dark")
```



High performers generally do get higher salary hikes, but the difference is not extreme. Some lower-rated employees still receive comparable hikes, suggesting that performance is a factor, but not the only one influencing pay increases.

8. What is the attrition rate in each department?(Which department loses the most employees?)

```
total_per_department = df['Department'].value_counts()
```

```
total_per_department
```

Department	count
R&D	961
Sales	446
HR	63

dtype: int64

```
left_per_department = df[df['Attrition'] == 'Yes']['Department'].value_counts()
left_per_department
```

Department	count
R&D	133
Sales	92
HR	12

dtype: int64

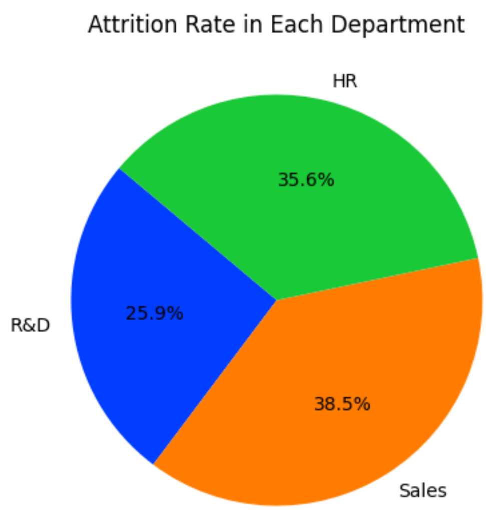
```
#calculate the attrition rate for each department now
attrition_rate = (left_per_department/total_per_department) * 10
```

```
attrition rate = attrition rate.round(2)
```

	count
R&D	13.84
Sales	20.63
HR	19.05

dtype: float64

```
plt.figure(figsize=(8, 5))
plt.pie(attrition_rate, labels=attrition_rate.index, autopct='%1.1f%%',startangle=140,colors=sns.color_palette("bright"))
plt.title('Attrition Rate in Each Department')
plt.show()
```



🏆 "Department-Wise Breakdown: Where Are We Losing Talent the Most?" This chart reveals the departments facing the highest attrition rates,

"Data-Driven Insights: Where Employee Turnover Hits Hardest"

Start coding or generate with AI.

9. Which Job Role has the highest average Percent Salary Hike?(Who's getting the best raises?)

Start coding or generate with AI.

Start coding or generate with AI.